Unit 1

Lesson 3

Spark The Challenge!

The Engineer's Journey.

Time Frame: [90 min]

Lesson Objectives:

By the end of this lesson students will be able to:

- Use the engineering design process to make a device that helps people talk using technology.
- Think of ideas that can help someone who cannot speak.
- Explain how your device helps people communicate.

Challenge questions of the lesson

1. **How can you use the steps of the engineering design process to create a device that helps someone who cannot speak communicate more easily?**
2. **What kind of technology could help someone who cannot speak share their thoughts, feelings, or needs?**
3. **How does your device work, and how would it make life easier for someone who cannot speak?`**

SEL Relation

- Responsible Decision-Making: Solving real-world problems with ethical thinking and innovation.
- Self-Management: Staying focused through a multi-step process to reach a goal.
- Relationship Skills: Collaborating with peers to design and improve solutions.
- Social Awareness: Showing empathy and understanding the needs of others.
- Self-Awareness: Recognizing personal values about helping others and creating inclusive solutions.

Engage

Teacher's Role:

- Begin the lesson by revisiting the goal: "How can we help Omar talk?"
- Read the story expressively to build excitement and curiosity.
- Prompt students with questions like:
    - o "What ideas do you have now that we know more about sound?"
    - o "How do engineers help people solve problems?"
- Guide the Career Readiness task by helping students imagine themselves as part of a team.
- Support students as they choose roles (Engineer, Programmer, Biomedical Engineer, Linguist).
- Encourage collaboration and sharing about their chosen job.

Expected From Students:

- Listen to the story and think critically about the challenge.
- Share possible ideas for how a talking device might work.
- Choose a career role and describe what they'd do to help the team.
- Work respectfully and contribute creative thinking.

Parts of the books included.


How can we help Omar talk?

Ms. Sara stands in front of the class and smiles.

Ms. Sara: "You told me you want to help Omar talk. Now that we know about sound, how can we do it?"

The class gets quiet. Everyone starts thinking.

Laila raises her hand.

Laila: "Maybe we can make something to help him talk?"

Adam leans forward.

Adam: "What if he writes, and a machine says the words out loud?"

Omar writes in his notebook:

"Yes! I want it to be fast and easy!"

Ms. Sara nods.

Ms. Sara: "Great ideas! Let's work like engineers and plan what to make."

STEAM SCOUTS 19

Pyramakerz®

Career Readiness

Imagine you're part of a team that is building a talking device to help people who cannot speak. Everyone has an important job. Choose your role, work with your group, and explain what you do to help the team!

Figure 12: A group of children working together to create a talking device.

The Engineers – "We Build the Device!"

Your Job:
You create the parts that make the device work. You connect wires, test batteries, and build the circuits so it can make sounds.

Figuer 13: Engineer.

20 STEAM SCOUTS

The Programmers – "We Teach It to Talk!"

Your Job:
You write simple code to tell the device what to say. You give it words and help it speak clearly.

Figure 14: Programmer.

The Linguists – "We Help It Sound Human!"

Your Job:
You help the device speak like a real person. You decide how fast it talks and what emotions it should show.

Figure 15: Linguist.

STEAM SCOUTS 21

Pyramakerz®

The Biomedical Engineers – "We Help People Use It!"

Your Job:
You work with people who need the device. You make sure it is easy for them to use and helps them talk with others.

Figure 16: Biomedical Engineers.

Explorer's Toolkit

Screen Lessons

Now it's time to step into the role of engineers and use the Engineering Design Process (EDP) worksheet to find a solution to the problem.
https://www.youtube.com/watch?v=MAhpfFt_mWM

22 STEAM SCOUTS

Story

- **Begin by gathering the class and setting a collaborative tone.**
- Say:
  - o “You’ve learned how sound works. Now, let’s see how we can use that knowledge to help someone speak—just like engineers do!”
- Read the story aloud using expression and pacing to emphasize curiosity and teamwork.
- **Prompt students with guiding questions like:**
  - o “What did Adam and Lila suggest?”
  - o “Why do you think Omar wants the machine to be fast and easy?”
  - o “What kind of help does Omar need from his friends?”

Possible Strategies for Implementation:

1. **Think–Pair–Share**
   - Ask: “What would your idea be to help Omar talk?”
   - Let students share with a partner before opening to the class.
2. **Emotion Cue Cards**
   - Show emoji-style cards and ask: “How does Omar feel when his friends try to help?”
   - Helps spark empathy and emotional connection.
3. **Picture Prompt**
   - Display a drawing of Omar holding a blank device. Ask: “What do you think should be on the screen or buttons?”
4. **Team Brainstorm Chart**
   - Use a chart labeled “Let’s Help Omar!” and let groups contribute device ideas (voice buttons, tablet, gesture sensors, etc.).

Career Readiness

Objective:

Students will explore different STEAM-related roles and responsibilities in a team working to design a talking device. They will begin to understand how collaboration and career specialization help solve real-life problems—especially in helping people with communication difficulties.

Possible Strategies for implementation:

1. **Role Assignment Game**
   - Present each role one at a time (Engineer, Programmer, Linguist, Biomedical Engineer).
   - Read the job description aloud and ask:
     - o “Who thinks they’d enjoy this job?” or
     - o “What kind of tasks do you think this job does every day?”
2. **Team Formation & Group Talk**
   - Divide students into groups and let each child choose a role from the team.
   - In groups, have them discuss:
     - o “What would your part of the device look like?”
     - o “What do you need to do your job?”
     - o “How will you help Omar?”
3. **Design Thinking Prompt Cards**
   - Give each team a card with questions tailored to their role (e.g., Engineers get a “What parts do we need?” prompt; Linguists get “How will the voice sound?”).
   - Let them sketch or write their ideas.
4. **Act It Out (Optional Drama Integration)**
   - Students perform short role-plays showing their character’s job.
   - Example: The Programmer types code on an invisible laptop and says “Hello!” in a robot voice.
5. **Reflection Exit Ticket**
   - Have students complete:
     - o “Today, I was a .... My job was to......... . I helped the team by ....”

Explain

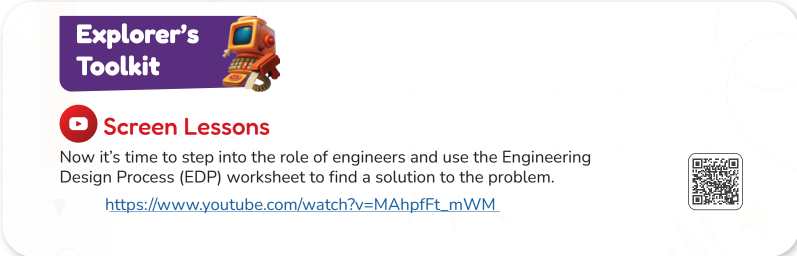
Teacher’s Role:

- o Facilitate discussion based on student ideas for a talking machine.
- o Encourage students to describe their “Ask” and “Imagine” steps out loud.
- o Help students fill in the “Plan” box on the EDP sheet with clear, simple steps.
- o Emphasize how each team role contributes to the design.

Expected from Students:

- Share their designs and thought processes.
- Describe what parts or features their talking device will include.
- Complete the "Plan" stage of the EDP using their team's input.

Parts of the books included.



Screen Lessons

Objective:

Students will understand the steps of the Engineering Design Process (EDP) and begin to think like engineers to solve communication challenges by creating a talking device.

Video link: https://www.youtube.com/watch?v=MAhpfFt_mWM

Possible Strategies for implementation:

1. **Pre-Watching: Set the Purpose**

- Ask:

  "What do engineers do when they want to build something?"

  "Have you ever had to plan something before making it?"

- Tell students:

  "In this video, we'll see how engineers solve problems step by step. Let's learn how to plan and build smart ideas—just like real inventors!"

2. **During Watching: Active Thinking**
   - Pause at key points and ask:
     - "What step are they doing now—asking, imagining, or planning?"
     - "What kind of problem are they trying to solve?"
   - Use a visual anchor (e.g., EDP chart poster) so students can point to the step being shown in the video.
3. **After Watching: Connect to Action**
   - Discuss:
     - "What was your favorite step?"
     - "What's the first thing we need to do to help Omar talk?"
   - Guide students to start filling in the Ask and Imagine sections of the EDP worksheet.

Story

- Continue reading the story where Adam, Lila, and Omar are ready to start building the talking device.
- Use expressive reading to bring characters to life, especially as they begin testing and improving.
- Emphasize the transition from planning to creating, linking it clearly to the Engineering Design Process.
- Pause to ask:
  - "What should they test first?"
  - "Why is it important to fix things that don't work?"
- Reinforce that mistakes are part of learning and real engineers improve their ideas.

Possible Strategies for Implementation:

1. **Role-Play & Build-Along**
   - Let students act out Adam, Lila, and Omar's next steps using classroom maker materials (mock buttons, cardboard, etc.).
   - Each group can choose parts to pretend to assemble—buttons, wires, a speaker, or a case.
2. **Think-Pair-Share**
   - Ask students: "What would you fix if something didn't work in your device?"
   - Let them discuss in pairs and share with the class.

3. **EDP Anchor Review**
   - Point to the "Create" step on the EDP chart. Ask:
     - "What does it mean to create something after planning it?"
     - "How will we know if our design works?"
4. **Use Visual Aids**
   - Show simple images of assistive devices or mock-up diagrams of the talking device.
   - Ask students to identify what parts might go inside the box and what they'll need to test.

Name: ....

Project: ....

Ask

What is the problem?

- Omar wants to talk but he cannot speak like others.
- How can we help Omar share his ideas using a device?

Imagine

What are some possible solutions?

- A device that says words when Omar writes or presses a button.
- A keyboard that speaks like Stephen Hawking's machine.
- A tablet that turns messages into sound.

Plan

What will we build? What will we need?

- Draw a sketch of a simple talking device.
- Make a list of needed parts: buttons, speaker, screen, battery, wires.
- Decide roles: Who will build, code, design the voice, or test the tool?
- Plan to make a case that covers all the electronic parts and makes the device easy to carry and protect.

Create

Improve

Elaborate

Apprentice

Objective:

Students will apply what they've learned about communication needs and sound technology to imagine and sketch their own version of a talking device that helps people who cannot speak.

Activity Instructions:

1. **Introduce the Task.**
   - Say: "Now that you've explored how sound works and learned what engineers do, it's time to become inventors!"
   - Explain that students will design a new talking device that could help someone like Omar communicate.
2. **Set the Scene.**
   - Remind students of the different jobs they explored in the Career Readiness section (e.g., Engineer, Programmer, Linguist, Biomedical Engineer).
   - Encourage them to think like a team and include helpful features in their design—buttons, speakers, display screens, lights, etc.
3. **Guide the Drawing Process.**
   - Ask guiding questions as they draw:
     - "Where will the sound come from?"
     - "How will someone use this device easily?"
     - "How will your design help them feel confident or understood?"
4. **Support All Learners.**
   - Allow students to label parts or add notes to their drawings if they're able.
   - Offer sentence starters like:
     - "This part helps because..."
     - "When someone presses this, it will..."
5. **Optional Extension.**
   - Pair students to share and explain their ideas with a peer.
   - Display the designs in a "Help Others Talk" wall gallery.

There is no right or wrong answer for this activity its left to students' imagination.

Now I Can

Objective:

Students will reflect on their learning by identifying personal achievements related to empathy, communication challenges, and how technology can help people who cannot speak.

Activity Instructions for Teachers:

1. **Introduce the Checklist:**

- Say: “Let’s think back on everything we learned. You’ve explored how to help someone who can’t speak by designing a talking device. Let’s check what we’ve learned together!”

2. **Guide Students Through Each Point:**

- Read each sentence aloud.
- Ask students to give a thumbs up if they feel confident, or simply check the box if they feel ready.
- Encourage them to explain why they checked a certain item using simple words, drawing, or gestures.

3. **Optional Extension:**

- Ask: “Which part of the project made you proud?”
- Invite them to circle one of the checkboxes and draw/write a sentence about what they enjoyed most or found the most helpful.

Relation to Standards

1. **English Language Arts (ELA):**

CCSS.ELA-LITERACY.SL.2.1

Participate in collaborative conversations with diverse partners about grade 2 topics and texts.

- Students discuss design ideas, challenges, and how to help others.

CCSS.ELA-LITERACY.SL.2.2

Recount or describe key ideas from a text or information presented orally or through other media.

- Students reflect on the story of Omar and Stephen Hawking to inspire their own ideas.

CCSS.ELA-LITERACY.W.2.8

Recall information from experiences or gather information from provided sources to answer a question.

- Students use what they learned from videos and the story to sketch and explain a solution.

2. **Next Generation Science Standards (NGSS):**

K–2-ETS1-1

Ask questions, make observations, and gather information about a situation people want to change.

- Students identify the challenge of helping someone who cannot speak and gather ideas to solve it.

K–2-ETS1-2

Develop a simple sketch, drawing, or physical model to illustrate how the shape of an object helps its function.

- Students draw their design for a talking device and describe its purpose.

K–2-ETS1-3

Analyze data from tests of two objects to compare the strengths and weaknesses of how each performs.

- (If students test or compare designs, this optional step can be included.)

3. **ISTE Standards for Students (Technology):**

Empowered Learner (1a, 1d):

- Students understand the basics of technology and how it can solve problems.
- They use engineering tools and processes to extend learning and build solutions.

Innovative Designer (4a, 4b):

- Students use a design process to generate ideas and create prototypes that help others.
- They plan and revise their ideas based on needs and feedback.